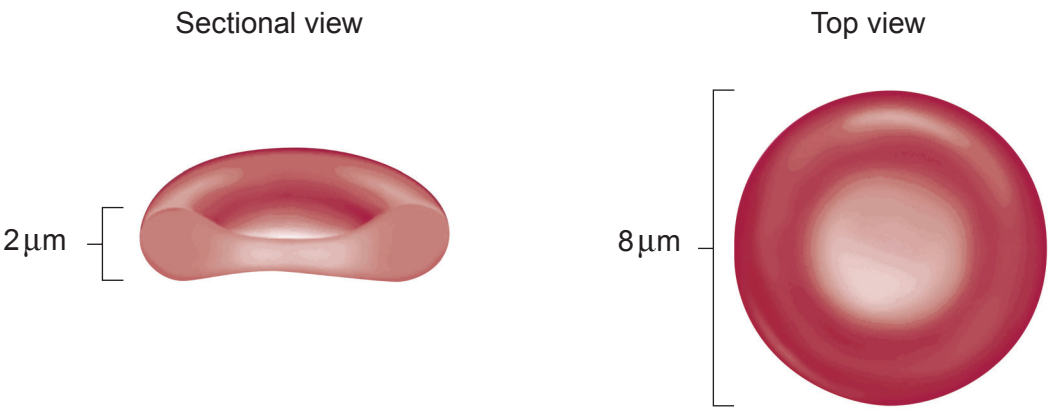


4. The main function of red blood cells (erythrocytes) is to carry oxygen from respiratory surfaces to respiring tissues.

The diagrams illustrate the structure of a red blood cell.



- (a) Describe and explain how the structure of a red blood cell is related to its function. [3]

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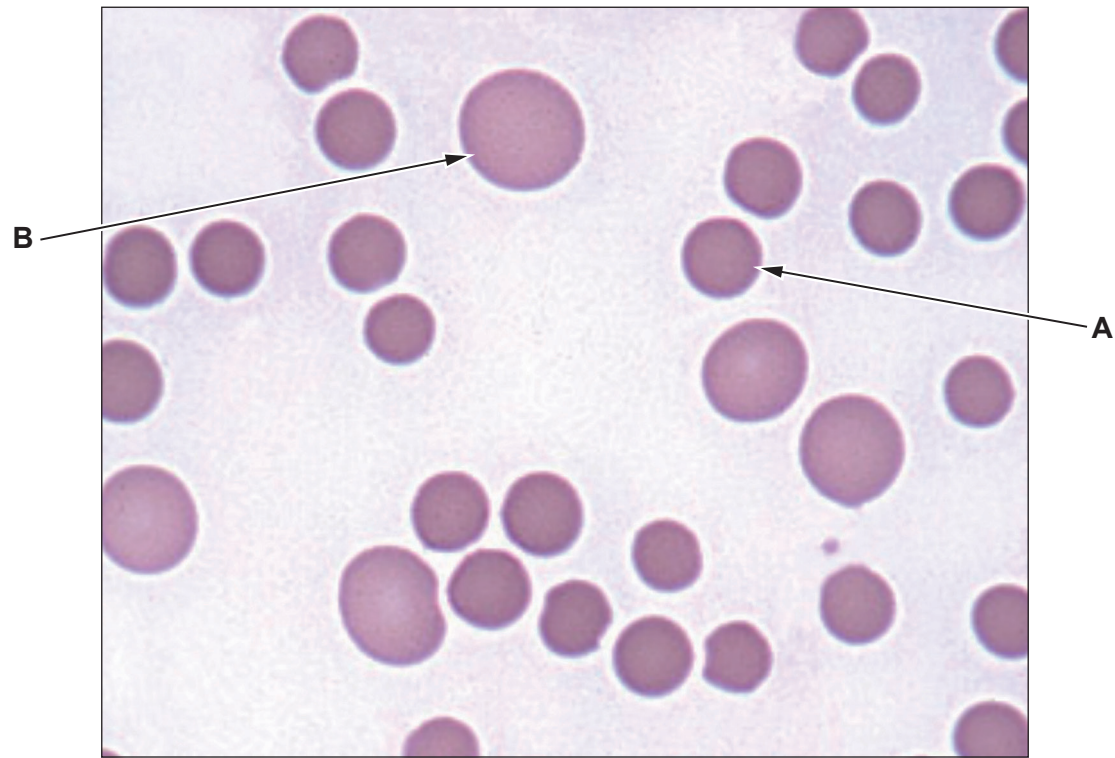
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(b) Anaemia may be defined as a decrease in the oxygen carrying capacity of the blood.

Macrocytic anaemia is caused by a dietary deficiency of vitamin B12 resulting in the production of fewer red blood cells than normal. Some of these red blood cells are very large and are called macrocytes.

The photomicrograph shows red blood cells of an individual with macrocytic anaemia.



Cell **A** is a normal red blood cell. It has a diameter of $8\mu\text{m}$. Cell **B** is a macrocyte. Calculate the diameter of cell **B to the nearest micrometre**. [3]

Diameter = μm



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- (c) A mean red blood cell volume that is higher than normal indicates macrocytic anaemia. The normal range is $80 - 95 \mu\text{m}^3$. The mean volume of a red blood cell in μm^3 can be calculated by using the following formula.

mean volume of a red blood cell = $\frac{\text{percentage volume of red blood cells in blood}}{\text{red blood cell count (cells per cubic millimetre)}} \times 10^7$

- (i) Calculate the mean volume of a red blood cell for a person with a red blood cell count of 4.5×10^6 cells per cubic millimetre where the percentage volume of red blood cells is 45%. [2]

Mean volume = μm^3

- (ii) State, with a reason, whether this person has macrocytic anaemia. [1]

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- (iii) Using the given information, suggest **two** reasons why macrocytic anaemia would cause a reduction in the oxygen carrying capacity of blood. Explain your answers. [2]

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